

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2196

Harrogate 138 kV
October 15, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2196. The Interconnection Customer proposes a 163.4 MW solar photovoltaic facility interconnecting at the Harrogate 138 kV substation in the Virginia area. The request was received on October 04, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 717 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$672,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Muscoda Industrial Substation – Bear Island 138 KV	Chancellor Row – Bear Island 345 KV	104.2	382	7.8
Dps Ocean Spray – Four Rivers 138 KV	Leavitt – Birchwood 345 KV	104.1	466	5.5
Trenton Diesel – Four Rivers 138 KV	Leavitt – Birchwood 345 KV Line	104.1	466	5.5
Walthall – Yellow Crk Substation 138 KV	Fritch – Big Hanaford 138 KV	101.2	354	23.5
Nstuca – Polk City 345 KV	Birchwood – Nauvoo 345 KV	101.1	1,458	5.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Muscoda Industrial Substation – Bear Island 138 KV	Reconductor line	\$1,570,000	6% (\$93,000)
Dps Ocean Spray – Four Rivers 138 KV line	Reconductor line	\$1,064,000	6% (\$69,000)
Trenton Diesel – Four Rivers 138 KV line	Reconductor line	\$4,307,000	6% (\$281,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Walthall – Yellow Crk Substation 138 KV	Reconductor line	\$1,301,000	7% (\$89,000)
Nstuca – Polk City 345 KV	Reconductor line	\$10,678,000	1% (\$140,000)
	Total allocated to project		\$672,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.